Question Paper

Exam Date & Time: 13-Oct-2020 (09:30 AM - 01:00 PM)

BMS COLLEGE OF ENGINEERING

Autonomous Institute Affiliated to VTU, Supplementary Semester End Examinations October 2020

Analysis and Design of Algorithms [15IS4DCADA]

Marks: 100 **Duration: 210 mins.**

Semester : IV - Information Science and Engineerin

Answer all the questions.

Instructions: 1. Answer FIVE full questions, using the given internal choices. 2. Missing data, if any, may be suitably assumed.

1)

a) Discuss the asymptotic notations with their definitions. (4)

b) Solve the following recurrence relations: (8)

(i) $T(n) = 2 * T(n/2) + 1$ with $T(1) = 0$

(ii) $T(n) = 2 * T(n-1) + 1$ with $T(1) = 0$

c) Derive the time complexity of the following Algorithms: (8)

(i) To find the nth Fibonacci number

(ii) Tower of Hanoi

2)

a) Show how the following numbers are sorted using quick sort with detailed steps. (6)

23 19 68 12 29 08 20 02

b) Apply Strassen's method to multiply the following 2 * 2 Matrix: (7)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{X} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Also, discuss how divide and conquer technique makes this method efficient.

c) Sketch and compare the Bubble Sort and Selection sort algorithms. Also, find the time complexity of both the algorithms. (7)

[OR]

3)

a) Apply Merge Sort for the following numbers. Show the Merge call tree with a neat Merge sort Algorithm. (10)

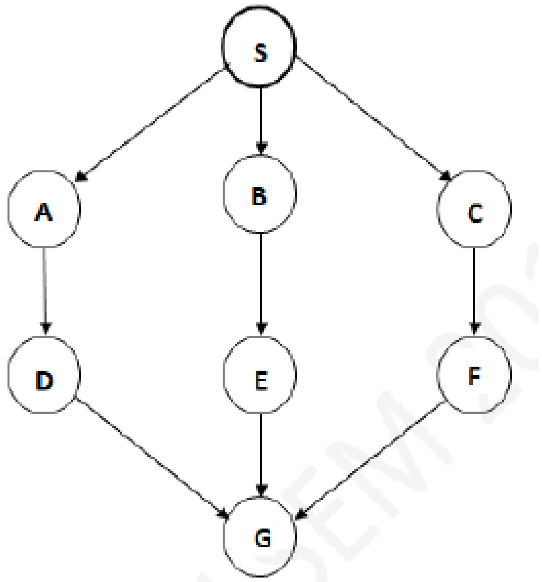
23 19 68 12 29 08 20 02

b) (5)

Demonstrate how Multiplication of large integers can be made efficient using Divide and conquer by multiplying 1234 and 5678.

c) Find the substring "KARNATAKA" in the string "BENGALURU IS THE CAPITAL OF KARNATAKA" using Brute force string matching method. Also, find the time complexity of this algorithm. . (5)

4) a) Apply BFS to traverse the following graph with the start vertex "S". Also, sketch the BFS algorithm. (8)



b) Sort the following elements using Insertion sort. Also, find the time complexity of Insertion Sort. (7)

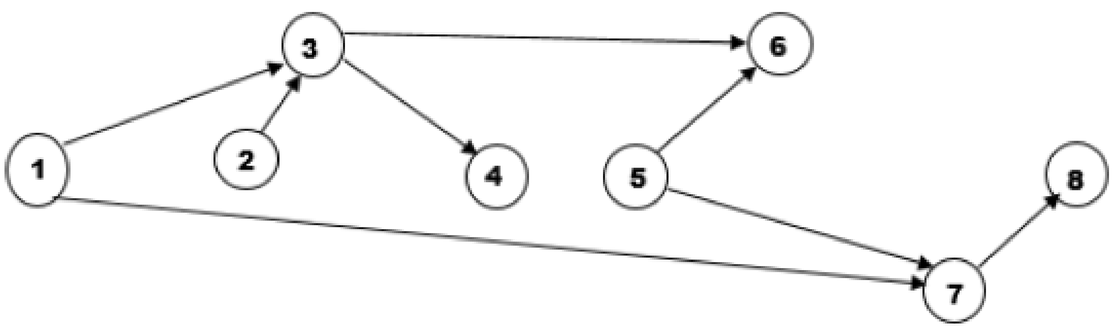
50 16 49 06 38 04 28

c) Construct AVL tree for the following elements in sequence. (5)

23 19 68 12 29 08 20 02 03 69

[OR]

5) a) Find the topological sequence for the following graph using DFS method starting from "1". (7)



b) Create Max Heap for the following elements. Also, show how the numbers get sorted using Heap Sort. (7)

25 98 34 02 56 45 34 04 92 12

c) Discuss the significance of pre-sorting. Also, construct a 2-3 tree for the following elements. (6)

23 19 68 12 29 08 20 02 03 69

6) a) Construct a Hash Table for the words: NEW DELHI IS THE CAPITAL OF INDIA AND IS LOCATED NORTH OF INDIA. Use Open Hashing by considering appropriate Hash table size and Hash function. (7)

b) Apply Dynamic Programming to solve the following Knapsack Problem. (7)

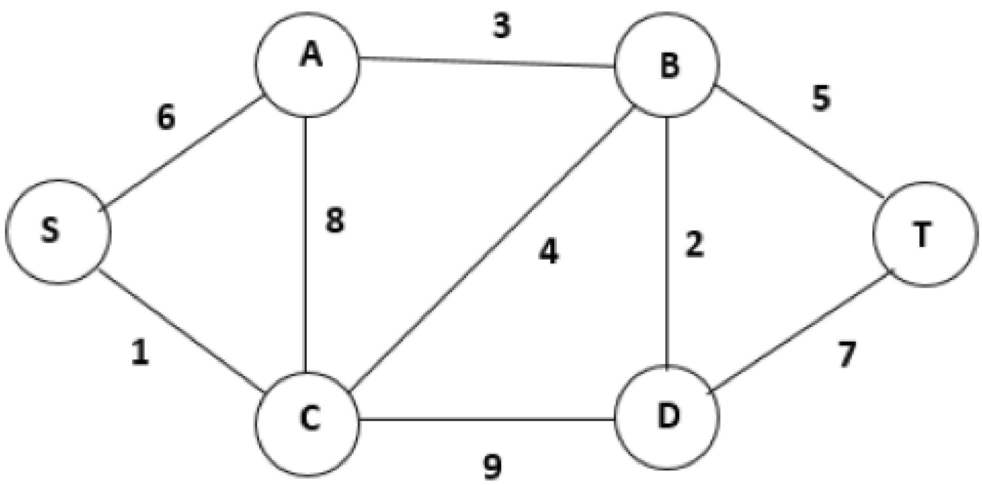
No. of objects n = 4, Maximum Weight W = 5,

Weight of each object = {2, 1, 3, 2}, Profit on each object = {12, 10, 20, 15}

c) Apply Horspool Algorithm to find the substring "SMILE" in the string (6)

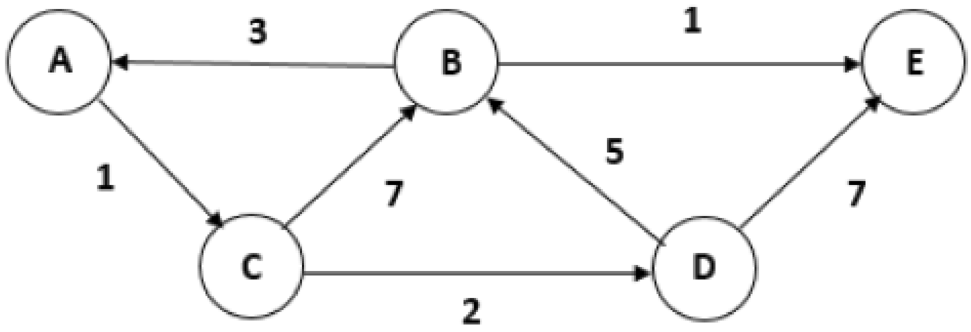
"THE RICHEST ORNAMENT FOR ANY PERSON IS A TRUE SMILE".

7) a) Apply Prim's algorithm for the following graph. Also, sketch the Prim's algorithm. (10)



b) (6)

Apply Dijkstra's algorithm for the following graph starting from "A".



c) Distinguish between P, NP and NP-Complete problems. (4)

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